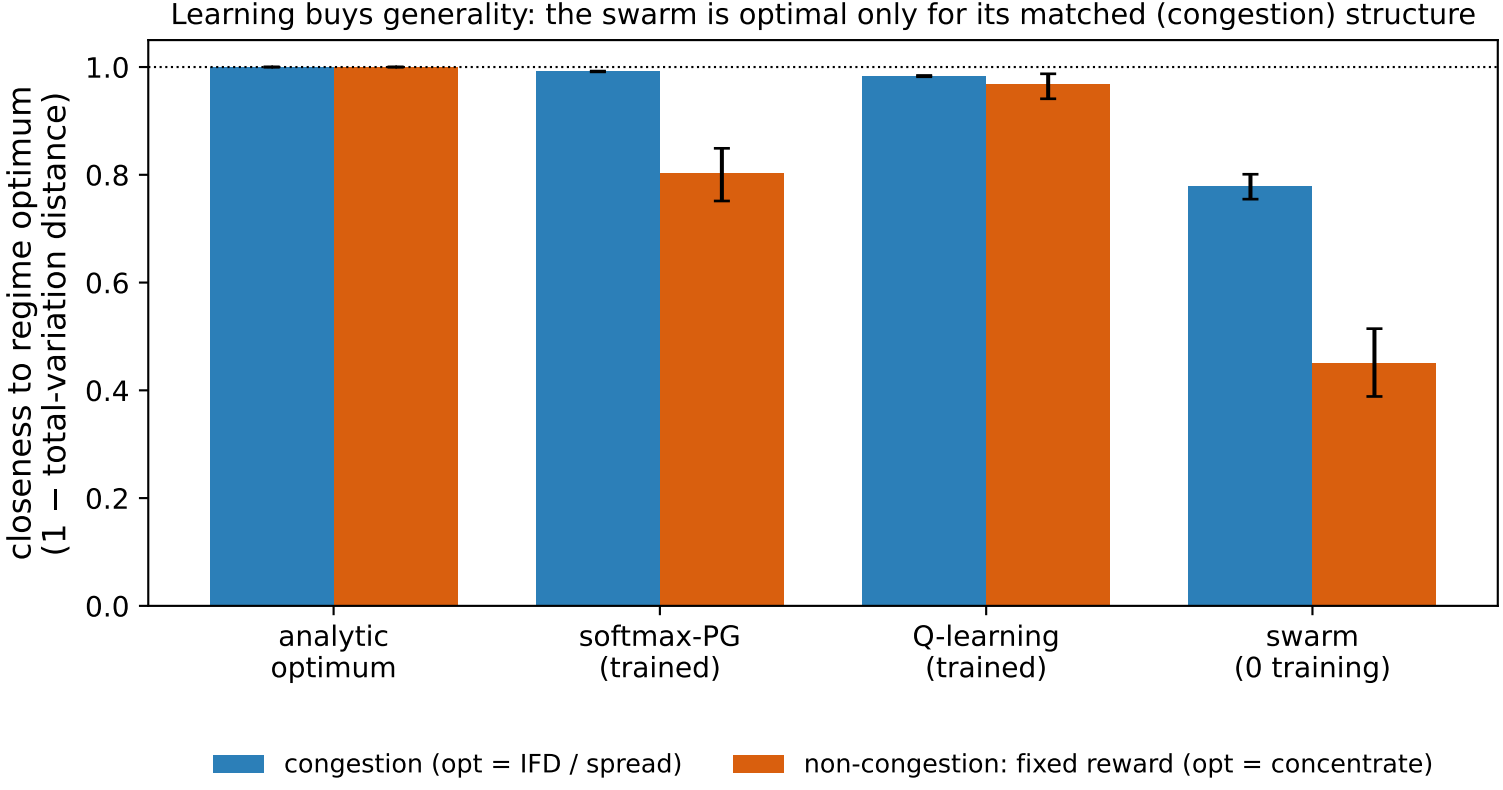
Learning buys generality: the swarm is optimal only for its matched (congestion) structure
closeness to regime optimum
(1 − total-variation distance)
1.0
0.8
0.6
0.4
0.2
0.0
analytic
optimum
softmax-PG
(trained)
Q-learning
(trained)
swarm
(0 training)
congestion (opt = IFD / spread)
non-congestion: fixed reward (opt = concentrate)